

Subject: Request for Five-Year Performance Analysis – Gold E-Commerce

Hi [Your Name],

As part of our upcoming international expansion strategy, we need a detailed analysis of Gold's business performance over the past five years. Gold is a fictional e-commerce company specializing in adventure and outdoor sports gear, offering nearly 300 products across the US, Canada, and Australia. We want to better understand how the business has evolved over time — what drives sales, how customer behavior is shifting, and which products contribute most to our growth.

We are providing you with five years of historical transactional data beginning in late 2010, along with complete datasets for 2011 through 2014, including the first month of each year. The dataset contains order activity, customer demographics, and product-level information. Your analysis should help us identify trends that can guide informed decisions as we prepare to enter new markets.

Your objectives for this analysis include:

- Examining how sales, orders, and customer volumes have changed over time
- Conducting cumulative performance analysis to measure long-term growth
- Assessing product categories and individual product contributions
- Identifying regional differences across the US, Canada, and Australia
- Segmenting customers based on purchasing behavior
- Providing data-driven recommendations to support market expansion

Tools to be used:

- SQL for all core analysis
- Excel for the visualization of insights

Specifically, we need insights on the following areas:

Changes Over Time Analysis

Identify revenue trends from 2010 to 2014. December appears to be a recurring peak, while 2012 shows a dip followed by strong recovery in 2013. Note any significant growth surges or declines — such as the sharp rise in late 2013 and the slowdown at the start of 2014.

Rolling Total Analysis

Evaluate how revenue accumulates annually. We're particularly interested in understanding the accelerating performance trend and the standout surge in 2013.

Moving Average Analysis

Assess how average product prices have evolved. The dataset indicates a consistent decline from 2011 to 2014, which may be driven by pricing strategies, competitive pressure, or category expansion.

Year-on-Year Analysis

Compare product-level yearly performance. Our preliminary understanding is that around 70% of product-year combinations show growth. Analyze the weaker 32% as well as the 104 instances that exceeded benchmark averages.

Part-to-Whole Category Analysis

Evaluate category-level contribution. Bikes currently dominate revenue, while clothing and accessories contribute minimally. Provide insight into diversification needs.

Product Segment Distribution

Break down product offerings by price segment. The catalog is heavily skewed toward budget and mid-range items, with a smaller share in high-end and premium categories. Explain what this means for customer acquisition and retention.

Revenue Share by Product Segment

Premium products, though few in number, generate 81% of total revenue. Analyze the implications for pricing, marketing, and inventory strategy.

Customer Distribution by Segment

Nearly 80% of customers are new. Only a small percentage convert into Regular or VIP customers. Investigate customer journey patterns and opportunities for retention improvement.

Revenue Distribution by Customer Segment

Regular and VIP customers contribute over 63% of total revenue despite representing only 20% of the customer base. We need recommendations on strengthening long-term loyalty and increasing customer lifetime value.

Please compile your findings using SQL for the analytical logic and Excel for all visualizations. Provide your final outputs as:

- SQL file containing all queries
- Excel workbook with graphs and supporting visuals
- Word document summarizing key insights and recommendations

Let me know if you need any additional information.

Regards,

[Executive Name]

Gold E-Commerce Strategy Team